Experiment - 1

Basic Networking Commands

Aim:

To study the various Network commands used in Linux of windows.

(1) arp - a ARP is short form of address resolution protocol, It will show the Ip address of your computer along with Ip address & MAC address of your routers.

Output :- gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3

(2) hostname : This is the simplest of all TCP/IP commands. It simply displays name of your computer.

output : localhost.localdomain

(3) netstat : (Network statistics) netstat displays a variety of statistics about a computer active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics e.t.c

eg :- net stat . -r

Output :

Destination	Gateway	Genmask	Flags	MSS	Window	irtt	Iface
default	-gateway	0.0.0.0	UG	0	0	0	enp0s
10.0.2.0	0.0.0.0	255.255.255.0	U	0	0	0	enp0s

(4) nslookup is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as ip address of particular computer, the MX records for a domain or the NS servers of a domain and non-interactive

eg: nslookup www.google.com

output:

Server: 8.8.8.8
Address: 8.8.8.8 #53
Non-authoritative answer:
Name: www.google.com
Address: 142.251.155.119

(5) ping (packet Internet Groper) command is the best way to test connectivity between two nodes ping use ICMP (Internet control Message protocol) to communicate no other devices.

output: pinging google.com [2404:6800:4007:838::200e] with 32 bytes of data.

(6) pathing: pathing is unique to window and it traces the route to destination address then launches 25 second test of each router.

output: Tracing route to google.com [2404:6800:4013:800::8b]. Over maximum of 30 hops.

(7) route: route command is used to show the IP routing table. It is primarily used to setup static routes to specific routes to specific routes host of networks via an interface.

O/p Interface List.

17 : ....00 00 27 00 00 11 ... Virtual box

Host – only ethernet adapter

IP V4 Route Table

Network destination | netmask | gateway | interface | metric

0.0.0.0 | 0.0.0.0 | 172.16.96.1 | 172.16.102.158

(8) ipconfig /all : This command displays detailed configuration information about your TCP/IP connection, Gateway, DNS, DHCP and type of ethernet adapter in your systems.

Op : enp0c3 : flags = 4163 <UP broadcast, Running 1500, inet 10.0.2.15 network 255.255.255.0 broadcast 10.0.2.55

lo : flags = 73 <UP, loopback, Running> netm 0xff36 inet 127.0.0.1 network 255.0.0.0 inet 6::1 prefixlen 128 Scopied 0x10<host

Some important Linux networking commands.

ip <options> a object d <command>

The ip command is one of the basic commands every administrator will need in daily work.

a. To show the IP address assigned to interface

a. [root@server ~]

o/p qdisc noqueue state unknown group
default qlen 1000 link/ loopback 00:00:00:00:00:00
brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host to valid_lft
forever preferred_lft.

b) ip route get.

Displays the route taken for IP 8.8.8. o/p
8.8.8.8 via 172.16.8.1 dev enp2s0 src
172.16.9.200 uid 3479.

c) ip link show - Displays all the network interfaces
and shows their status.

o/p 1: lo <LOOPBACK, UP, LOWER_UP> mtu 65536
qdisc state unknown mode DEFAULT

d) ip route show - Displays system routing table including
Default gateway, network routes.

Output: default via 172.16.8.1 dev enp2s0 proto
static metric 100.

2. ifconfig: This command is a state in many
system, tool belt configuring troubleshooting
networks. It has been replaced by ip command
above

Output: flag: 4163 UP, BROADCAST, RUNNING, multicast
mtu 1500 inet 172.16.9.200 net mask 255.255.255.0
broadcast 172.16.11.255

(3) mtr - Matt's traceroute is a program with a command like interface that serves as a network diagnostic troubleshooting Just like trace route mtr command mtr provides a lot of statistics about each such as response time, percentage with mtr, you will get more information about route be able to see problematic devices along the way O/p:

fedora (172.16.9.200) → google.com
(142.250.77.110) 2026-07-23 01:44:36-040

keys: Help display mode Restart statistics order of fields.

Host	Loss %	Snt	Last	Avg	Best	Wrst	StDev
gateway	14.7%	200	0.4	0.5	0.2	14.5	2.4

(b) show numeric IP address - mtr -g. google.com
output: fedora (172.16.9.200) → google.com
(142.250.67.46) 2026-07-2376

keys: Helps Documentation Restart statistics
order of fields.

Host	Packets	Snt	Last	Avg	Best	Wrst
172.16.8.1	0.0%	107	0.3	0.4	4.9	0.5

(c) Show numeric IP with host name mtr -b google.com
output:

Host	Packets	Snt	Last	Avg	Best	Wrst	Dev
gateway	0.01 %	56	6.2	2.7	0.2	12.9	5.3
mail (182.156.9.89)	13.8	84	26.2	21.6	1.8	31.5	4.2

d) set the number of pings that you want to send.

output :-

host	packets	snt	avg	best	worst
	cust%				
localhost	0.0%	7	0.1	0.0	0.1

4) tcpdump : is designed for capturing, displaying packets

output : [$ sudo] password for Linux : root.

5) ping

ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet control Message protocol (ICMP) Echo request Messages.

you need to ping stop the command by pressing CTRL + c. otherwise, it will ping until you stop it.

If you want to ping a host ten times, use the following command :

[root@server ~] # ping -c 10 google.com

Experiment-2

Aim: study of different types of Network cables

a) Understand types of network cables

Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted pair (STP) cable
3. Coaxial cable
4. fibre optic cable

Cable Type	category	Maximum Data Transmission	Adv/ Disadv	Application & use
UTP	category 3	10 bps	Adv • cheaper in cost • easy to install Disadv • More prone to (EMI)	10 Base T-Ethernet
	category 5	up to 100 Mbps		Fast Ethernet
	category 5e	1 Gbps		
STP	category 6, 6a	10 Gbps	Adv • shielded • faster than UTP • Less susceptible to noise and interference	Gigabit Ethernet 10 G Ethernet (55m)

STP	category 7	10 Gbps	disadv o Expensive o Greater installation effort	Gigabit Ethernet, 10G Ethernet (100m)
Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	adv o High bandwidth o Low loss Bandwidth o Immune to interference o versatile Disadv o Limited distance o cost o size is bulky	Speed of signal is 500m Television network High speed internet connections
fibre optics cable	Single mode multi-mode	100Gbps	adv o High speed o High bandwidth o High security o Long distance Disadv o Expensive o requires skilled installers.	o maximum distance of fibre optics cable is around 100 meters

b) Make your own Ethernet cross-over cable / straight cable

Tools and parts needed:

- Ethernet cabling cat5e is certified for gigabit support, but cat5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs
- optional two plug shields.

Step 1 :- To start construction of the device, begin by threading shields onto the cable

Step 2 :- Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3 :- After, you will need to untangle the wires, there should be four "twisted pairs." Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4 :- Once the order is corrected, bunch them together in a line, and if there are any that stick out farther than others, slip them back to create an even level. The difficult aspect is placing them into the RJ45 plug without messing up the order. To do so hold the plug with the clip side facing away from you and have the gold pins facing toward you.

Step 5 :- Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't, that means that you stripped off too much shielding. simply slip the cables back a little more.

Step 6 :- After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be shaped correctly.

Step 7 :- Lastly, repeat for the other end using diagram B (to make a crossover cables) using diagram A. (to make a straight through cable).

Experiment - 3

Aim :-

To study the packet tracer tool Installation and User Interface overview.

c) To understand environment of CISCO PACKET TRACER to design simple network.

INSTALLING PACKET TRACER :-

To download packet Tracer, go to https://www.netacad.com and log in with your Cisco Networking Academy Credentials.

windows

Installation in windows is simple. the setup comes in single file named packettracer_setup6.0.1.exe. accept the license agreement, choose a location and start the installation

LINUX

Linux users with an Ubuntu/Debian distribution should download the file for Ubuntu, and those using fedora/redhat/centos must download the file for fedora.

chmod +X PacketTracer601_i386_installer-rpm.bin
./PacketTracer601_i386_installer-rpm.bin

d) Analyse the behaviour of network devices using CISCO PACKET TRACER simulator.

1. From the network component box, Click and drag-and-drop the below components :
   a. 4 Generic PCs and one HUB
   b. 4 Generic PCs and one switch

2. Click on connections:

a. Click on copper straight - Through cable

b. select one of the PC and connect it to HUB using the cable. The link LED should glow in green, indicating that the link is up

c. Similarly connect 4 PCs to the switch using copper straight - Through cable.

3. Click on the PCs connected to hub, go to the Desktop tab, click on IP configuration and enter an IP address and subnet mask.

4. Observe the flow of PDU from source PC to destination PC by selecting the real time mode of simulation

5. Repeat step #3 to step #5 for the PC's connected to the switch

6. Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about behaviours of switch and HUB.

Experiment - 4

Aim:

Setup and configure a LAN using a switch and ethernet cables in your Lab

What is a LAN?

It refers to a network that connects devices within a limited area. such as office building, school or home. It enables users to share resources, including data, printers and internet access. LAN connects devices to promote collaboration and transfer between users, such as computers, printers, servers and switches. A LAN switch serves as the primary connecting device, managing and directing communications with the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

How to setup a LAN?

step-1: plan & design an appropriate network topology taking into account network requirements and equipment location.

step-2: You can take four computers, a switch with 8, 16, or 24 ports which is sufficient for networks of these sizes and four ethernet cables.

Step-3: Connect your computers to network switch via an ethernet cable, which is simple as plugging one end of the ethernet cable into your computer and the other end into your network switch.

Step-4: Assign IP address to your PC's.

(1) Log on to the client computer as Administrator or as owner.
(2) Click network and internet connections.
(3) Right click local area connection / Ethernet → Goto properties → Select internet protocol (TCP/IPv4) → Click on properties → Select use the following IP address option and assign it

Similarly assign Ip address to all the pcs connected to switch. PC1 - Ip address: 10.1.1.1, subnet mask 255.0.0.0
PC2 - Ip address - 10.1.1.2, subnet mask 255.0.0.0 PC3 - Ip address 10.1.1.3, Subnet mask 255.0.0.0.
PC4 - Ip address 10.1.1.4, Subnet mask 255.0.0.0

Step-5: Configure a Network Switch.

(1) Connect your Computer to the Switch: To Access the switch's web interface need to connect your Computer to the Switch. using an ethernet cable.

'og in to the web interface.

(c) Configure the basic settings.

(d) Assign Ip address.

Step-6:

Check the connectivity between Switch and other machine by using ping in the command prompt of the device

Step-7: Select a folder → Goto properties → Click Sharing tab → Share it with every same lan.

Step-8: Try to access the shared folder from other computers of the network.

Experiment - 5

AIM :

Experiments on packet capture tool : wireshark

Packet sniffer

- sniffs messages being sent / recieved from / by your computer
- store and display the contents of the various protocol fields in the messages
- passive program
  - never sends packets itself
  - no packets addressed to it
  - recieves a copy of all packets (sent / recieved)

Packet sniffer Structure Diagnostic Tools

- Tcpdump
  - E.g. tcpdump -enx host 10.129.41.2 -w exe3.out
- wireshark
  - wireshark -r exe3.out

Description :

What we can do with wireshark :

- capture network traffic
- Decode packet protocols using dissectors
- Define filters - capture and display
- watch smart statistics

- Analyze problems
- Interactively browse that traffic

wireshark used for :

- Network administrators : Troubleshoot network problems
- Network security engineers : examine security problems
- Developers : debug protocol implementations
- people : learn network protocol internals.

Getting wireshark

wireshark can be downloaded for windows or macOS from it's official website.

Capturing packets

After downloading and installing wireshark, launch it and double-click the name of a network.

THE "Packet List" pane

Displays all the packets in the current capture file.

THE "PACKET DETAILS" pane

The packet details pane shows the current packet (selected in the "packet list" pane) in a more detailed form.

THE "PACKET BYTES" pane

The packet bytes pane shows the data of the current packet in a hexdump style.

Color Coding

Wireshark uses colors to help identify the types of traffic at a glance.

To view exactly what color codes mean, click View > Coloring Rules.

filtering packets

You have a large amount of packets to sift through. That's where Wireshark's filters come in. The most basic way to apply a filter is by typing it into the filter box at the top of the window and clicking Apply.

You can also click Analyze > Display filters to choose a filter from among the default filters included in Wireshark.

Inspecting packets

Click a packet to select it and you can dig down to view it's details.

CAPTURING AND ANALYSING PACKETS USING WIRESHARK TOOL

To filter, capture, view packets in wireshark Tool. capture 100 packets from the Ethernet : IEEE 802.3 LAN Interface and save it.

procedure :-

- select local Area connection in wireshark
- Go to capture option
- select stop capture automatically after 100 packets.
- Then click start capture
- Save the packets

Experiment-6

AIM :-

write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature.

Error correction at Data Link layer:

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the receiver. It is a technique developed by R.W. Hamming for error correction.

create sender program with below features

1. Input to sender file should be a text of any length. program should convert the text to binary
2. Apply hamming code concept to the binary data and add.

CODE:

```
def calc-redundant_bits(m):
        for r in range(m+1):
            if (2**r) >= (m+r+1):
                return r
```

```
def main():
    data = input("Enter data bits: ").strip()
    m = len(data)
    r = calc_redundant_bits(m)
    n = m + r
    print(f"Number of r needed: {r}")
    print(f"Total frame length: {n}")
    code = ['0'] * (n+1)
    j = 0
    for i in range(1, n+1):
        if (i & (i-1)) != 0:
            code[i] = data[-(j+1)]
            j += 1
    for i in range(r):
        parity_pos = 2**i
        val = 0
        for pos in range(1, n+1):
            if (pos & parity_pos) == 0 and pos != parity_pos:
                if code[pos] == '1':
                    val ^= 1
        code[parity_pos] = str(val)
```

```
transmitted_str = "".join(code[::-1])
print(f"Data to be transmitted: {transmitted_str}")

change = input("\n Transmitted bit change").strip().upper()

if change == "YES":
    received_str = input(f"Enter corrupted data").strip()
    if (n(received_str) != n):
        print(f"Error: Received data must be exactly {n} bits long!")
        return
    received_code = ['0'] + list(received_str[::-1])
else:
    received_code = list(code)

error_position = 0
for i in range(r):
    parity_pos = 2**i
    val = 0
    for pos in range(1, n+1):
        if (pos & parity_pos) == 0:
            if received_code[pos] == '1':
                val ^= 1
    if val != 0:
        error_position += parity_pos
if error_position == 0:
    print("\n verification result: No error detected")
else:
    print("\n verification result: Error detected:")
```

```
        print(f"wrong bit at position: {error_position}")
        received_code[error_position] = '0' if received_code[error_position] == '1' else '1'
    corrected_str = ''.join(received_code[1:][::-1])
    print(f"corrected data stream: {corrected_str}")

if __name__ == "__main__":
    main()
```

OUTPUT :

① enter data bits : 101100
Number of redundant bits (r) needed : 4
Total frame length (m+r) : 10
Data to be transmitted : 1011100001
transmitted bit change ? (Yes/NO) : NO
Verification Result : NO error detected in transmission.

RESULT :- Thus the Hamming code was implemented successfully for error detection and correction.

Experiment - 7

AIM :-

write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. simulate the flow of frames from one node to another.

Create a sender program with following features :-

1. Input window size from the User.
2. Input a text message from the user.
3. Consider 1 character per frame
4. Create a frame with following fields [frame no., DATA].
5. Send the frames [print the output on screen]
6. Wait for the acknowledgement from the Reciever.
7. Reader a file called Recievers_Buffer.
8. check ACK field for the ACKnowledgment number.
9. If the acknowledgement number is as expected, send new set of frames accordingly, Else if NACK is recieved, resend the frames accordingly.

create a reciever file with following features

1. Reader a file called Senders_Buffer.

2. check the frame no.
3. If the fame no. are as expected, write the appropriate Ack no. in the Reciever_Buffer file. Else write NACK no. in the Reciever_Buffer file.

CODE

```
import time

message = input ("Enter the text message: ")
window_size = int (input ("Enter a window
                          size : "))
error_frame = int (input ("Enter error frame
                          (0 for no error): "))


if window_size <= 0:
    print ("Invalid window size")
    exit ()

print ("\n")
print ("SLIDING WINDOW PROTOCOL")

print ("\nMessage :", message)
print ("window size :", window_size)

start = 0
expected_frame = 1
error_included = False

while start < len (message):

end = min (start + window_size, len (message))
```

```
print("\n")
print(" SENDER BUFFER")

sender_buffer = []
for i in range(start, end):
    frame_no = i + 1
    data = message[i]
    sender_buffer.append([frame_no, data])
    print("FRAME_NO = ", frame_no, "Data = ", data)
    print("\n sending frames : ")
    time.sleep(1)

print("\n")
print("RECEIVER")
error_found = False
for frame in sender_buffer:
    frame_no = frame[0]
    data = frame[1]
    if frame_no == error_frame_not and Error induced:
        recieved_frame = frame_not + 1
        error_induced = True
    else:
        recieved_frame = frame_no
    print(
        "Recieved frame : ",
        recieved_frame,
        "DATA : ",
        data
    )
```

```
if recieved_frame == expected_frame:
        print("frame", recieved_fram,
        "is correct")
        expected_frame += 1

else:
        print("\n** ERROR DETECTED **")
        print("Expected frame :",
              expected_frame)
        print("Recieved frame :",
              recieved_frame)

print("\n")
print("  RECIEVER BUFFER")
        print("NACK =", expected_frame)
        print("\n Nack recieved by sender")
        print("Resending frames from
              frame", expected frame)

        error_found = True
if error_found:
        break
        start = expected_frame - 1
        time.sleep(1)
        continue

print("\n")
print("RECIEVER BUFFER")
```

```
ack = expected_frame
print("Ack = ", ack)

print("\n Ack recieved by Sender")
print("Expected Ack : ", end+1)

if ack == end+1 :
        print("\n Ack is correct")
        print("sliding window moves
              forward")
        start = end

else :
        print("\n Incorrect Ack")
        print("waiting for correct
              Ack")

print("\n")
print("TRANSMISSION COMPLETED")

print("\n Original message : ", message)
print("All frames successfully transmitted.")
```

OUTPUT :-

```
Enter the text message : HELLO
Enter window size : 3
Enter error frame ( 0 for no error ) : 0
```

SLIDING WINDOW PROTOCOL

Message : HELLO
window size : 3

SENDER BUFFER

FRAME_NO = 1 DATA = H

FRAME_NO = 2 DATA = E

FRAME_NO = 3 DATA = L

RECEIVER

Recieved frame : 1 DATA : H
Frame 1 is correct
Recieved frame : 2 DATA : E
frame 2 is correct
Recieved frame : 3 DATA : L
frame 3 is correct

ACK = 4
sliding window moves forward

SENDER BUFFER
FRAME_NO = 4 DATA = L
FRAME_NO = 5 DATA = O

RECIEVER
frame 4 is correct
frame 5 is correct

ACK = 6
sliding window moves forward
Transmission completed

Result :-
All frames successfully transmitted

Experiment - 8

AIM:

To Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform

Room Link : https://tryhackme.com/room/nmap01

Introduction

When targeting a network, we need an efficient tool to handle repetitive tasks. This tool should help us find out which systems are active and what services are running on those systems. This room is the first in a series of four rooms dedicated to Nmap rooms that focus on port-scanning.

This room is the first of four in this Nmap series. These four rooms are also part of the Network Security module.

- Nmap Live Host Discovery
- Nmap Basic port scans
- Nmap Advanced port scans
- Nmap post port scans

We present the different approaches that Nmap uses to discover live hosts. In particular we cover :

ARP Scan : This scan uses ARP requests to discover live hosts.

ICMP scan : This scan uses ARP requests to discover live hosts.

TCP/UDP ping scan : This scan sends packets to TCP ports and UDP ports to determine live hosts.

Subnetworks

A network segment is a group of computers linked through a shared medium, like an ethernet switch or Wi-Fi access point. In IP networks, a subnetwork typically consists of one or more network segments connected together and configured to use the same routes. A network segment is a physical connection, while a subnetwork is a logical connection.

the figure displays two types of subnets:

/16 subnets: These have a subnet mask of 255.255.0.0 and can accomodate approximately 65 thousand hosts.

/24 subnets: These feature a subnet mask of 255.255.255.0 and can support around 250 hosts.

- from computer 1
- To computer 1 (to indicate it is broadcast)
- packet Type: "ARP Request"
- Data: computer 6 (because we are asking for computer 6 MAC address using ARP Request)

NMAP Host Discovery using ARP

Nmap employs the following strategies to find live hosts:-

When a privileged user tries to scan targets on a local network (Ethernet), Nmap uses ARP requests. A privileged user is root or a user who belongs to sudoers and can run sudo.

when a privileged user tries to scan targets outside the local network, Nmap uses ICMP echo requests, TCP ACK to port 80, TCP SYN to port 443 and ICMP timestamp request.

when an unprivileged user tries to scan targets outside the local network, Nmap resorts to a TCP 3-way handshake by sending SYN packets to port 80 and 443.

UDP ping

you can also employ UDP to check if the host is online. Unlike TCP SYN ping, sending a UDP packet to an open port typically doesn't elicit a response.

However, when sending a UDP to a closed UDP port you anticipate receiving an ICMP "port unreachable" packet, which indicates the target system is active and reachable

while sending UDP packets to open UDP ports may not trigger a response, sending them to closed UDP ports.

EXPERIMENT – 8 B)

AIM :-

configuration of wireless LAN using cisco packet Tracer.

Design a Topology with three PCs connected from Linksys wireless routers.

perform following configuration :-

- configure static IP on PC and wireless Router
- set SSID to MotherNetwork
- set IP address of router to 192.168.0.1, PC0 to 192.168.0.2, PC1 to 192.168.0.3 and PC2 to 192.168.0.4.
- secure your network by configuring WAP key on Router
- connect PC by using WAP key

To complete these tasks follow these step by step instructions:-

Step 1 :- Click on wireless router,

- select administration tab from top menu, set Username and password to admin and click on save setting.
- Next click on wireless tab and set default SSID to MothaNetwork.
- Now select wireless security and change security Mode to WEP
- Set key1 to 0123456789
- again go in the end page and click on Save setting
- Now we have completed all given task on Wireless router. Now configure the static IP on all three PC's

- Double click on PC select Desktop tab click on IP configuration select static IP and set IP.
- connect PC's from wireless routers. To do so click PC select Desktop click on PC wireless.
- click on connect tab and click on Refresh button.
- Repeat same process on PC1 and PC2.

Experiment -9

AIM :-

Implementation of subnetting in CISCO PACKET TRACER simulator.

Classless IP subnetting is a technique that allows for more efficient use of IP addresses by allowing for subnet masks that are not just the default masks for each IP class.

CREATING A NETWORK TOPOLOGY

The first step in implementing classless IP subnetting is to create a network topology in packet Tracer. To create a network topology in packet Tracer, select the "New" button in the top left corner, then select "network" and "Generic".

ADDING THE DEVICES :

Once we have created our network topology, we can add devices to it. To add a device from the bottom left corner and drag it onto the network topology.

SUBNETTING:

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch, and the routers, we can use a 24+7 subnet mask. This will give us 8 subnets with 30 host addresses each.